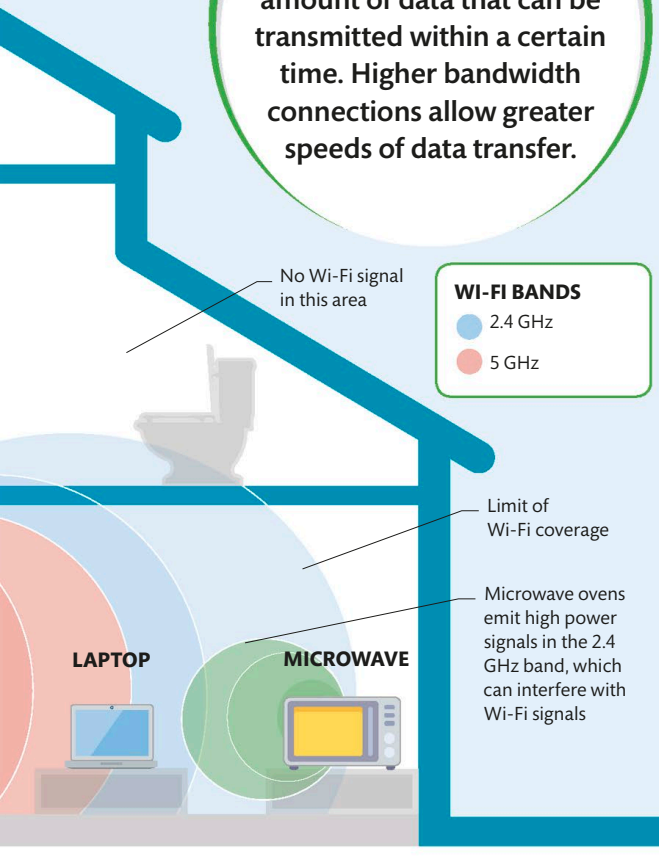
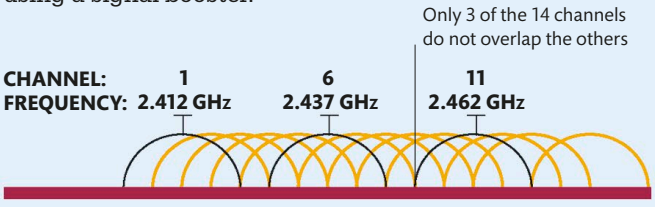
WHAT IS BANDWIDTH?

Bandwidth refers to the amount of data that can be transmitted within a certain time. Higher bandwidth connections allow greater speeds of data transfer.

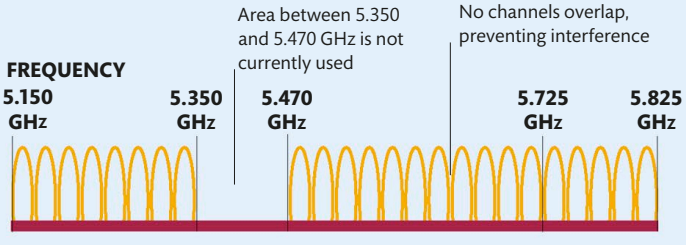


Wi-Fi signals

The strength of a Wi-Fi signal falls rapidly with the distance between device and router. Wi-Fi range is usually up to 150 feet but can vary depending on frequency, transmission power, and antennas. Range tends to be smaller indoors due to the presence of obstacles, such as walls, although it can be extended using a signal booster.



2.4 GHz spectrum
Data is transmitted using specific frequencies (channels) that multiple devices can share. Using multiple channels enables more efficient communication, but in the 2.4 GHz band (shown here) many channels overlap, causing interference.



5 GHz spectrum
The 5 GHz spectrum has 24 nonoverlapping channels, making use of higher frequencies. This means that data can be transmitted through many channels simultaneously for greater efficiency and speed. Wi-Fi systems can use the area of the spectrum from 5.725–5.875 GHz but only for short-range, low-power devices.

HACKING WI-FI

Wireless Internet connections are vulnerable to hacking because a hacker can access a Wi-Fi network without having to be in the same building or needing to break through a firewall. Hackers can breach Wi-Fi security in various ways, including harvesting information transmitted and received by devices. A wireless network can be secured with Wi-Fi Protected Access. This relies on the user entering a verified password and works by producing new encryption keys for each data packet.

